

Guarded Block Transfers for Polytope Distance: Linear Convergence with Explicit Constants and Gap-Certified Screening for SVM Training

author names withheld

Under Review for OPT 2026

Abstract

Hard-margin SVM training is the distance problem between two convex hulls, which the Triangle Algorithm (TA) solves with a certified duality gap. We propose *guarded block transfers*: k Mitchell–Demyanov–Malozemov (MDM) transfers per scan are aggregated into one exact line search, guarded by a comparison with the single best transfer at a cost independent of the number of points n , and replaced by an away step when the leading transfer is capacity-clipped. On the Minkowski-difference polytope, where the objective is 1-strongly convex, we prove global linear convergence with explicit constants and no combinatorial swap-step bounds. Since the TA certificate is the Frank–Wolfe gap divided by the current distance, gap-safe screening applies, and we bound the iteration after which it has removed every non-support point by $O\left(\frac{k}{\rho} \log \frac{1}{\tau}\right)$, ρ the rate constant and τ the optimal-face margin. Every inequality has a numerical check. On five LIBSVM benchmarks the solver matches the baselines to five digits wherever both converge, wins or ties against LIBSVM and kernel SMO on sparse-support cells, and is one to two orders of magnitude slower on dense-support L_2 margins and nearly touching kernel hulls, where LIBLINEAR finishes in seconds. We report both sides.

1. Introduction

Training a linear hard-margin SVM is equivalent to finding the closest points of the two classes’ convex hulls: the optimal separator is the perpendicular bisector of the witness pair and the margin is half the hull distance [8]. Kalantari’s Triangle Algorithm (TA) exploits this *distance duality* directly [1]: TA I decides whether the hulls intersect, and TA II computes their distance while maintaining an iterate pair (p, q) and a certified interval $[LB, UB]$ from parallel supporting hyperplanes. The enhanced TA (ETA) of [2] adds dot-product caching and search heuristics.

The workhorse step of such solvers is the *weight transfer* from the worst active vertex to the best candidate, the Mitchell–Demyanov–Malozemov (MDM) step [4], which is the pairwise Frank–Wolfe (FW) step of [3] on the relevant polytope. Each step costs an $O(n)$ scan and yields one transfer; we ask how much of the scan can be reused across several transfers without weakening the guarantee.

Contributions.

1. **Guarded block transfers** (§3): pair the top- k receivers with the worst- k active donors, take one exact line search along the aggregated direction, *guard* it by an $O(k^2)$ comparison with the single MDM step, and fall back to an away step on capacity-clipped iterations.

2. **Linear convergence with explicit constants** (Theorem 3): on $Z = \text{conv}(V) - \text{conv}(W)$ the guarded block algorithm inherits the pyramidal-width rate of pairwise FW up to a constant, and the away fallback replaces the combinatorial swap-step bound by linear drop-step counting (cf. [12]).
3. **Screening activation with explicit constants** (Theorem 6): the TA certificate is the FW gap divided by the distance (Lemma 5), so gap-safe screening [16] applies, and we bound the iteration by which it has removed every zero-weight non-support point.
4. **A regime study on real data** (§5): the analysis predicts $O(n^2)$ -type cost on dense supports (pyramidal width $\sim 1/\sqrt{n}$), and two batteries on five LIBSVM datasets show the boundary from both sides.

Code, proofs, numerical checks and experiment scripts are in the anonymised supplementary repository.

2. Setting

Let $V = \{v_1, \dots, v_n\}$, $W = \{w_1, \dots, w_m\} \subset \mathbb{R}^d$. The polytope distance problem is

$$\min_{p \in \text{conv}(V), q \in \text{conv}(W)} \frac{1}{2} \|p - q\|^2. \quad (1)$$

Writing $z = p - q$ and $Z := \text{conv}(V) - \text{conv}(W) = \text{conv}\{v_i - w_j\}$, (1) is the minimum-norm-point problem $\min_{z \in Z} F(z)$, $F(z) = \frac{1}{2} \|z\|^2$, whose objective is 1-strongly convex and 1-smooth. Let z^* be the minimiser, $\delta^* = \|z^*\|$, $h(z) = F(z) - F(z^*)$, $M = \text{diam}(Z)$ and $\delta = \text{PW}(A) > 0$ the pyramidal width [3] of the atom set $A = \{v_i - w_j\}$ (a property of the atom set, not of Z alone).

The algorithm keeps simplex weights α on V , β on W (iterate z_t after t iterations), with $p = \sum \alpha_i v_i$, $q = \sum \beta_j w_j$; the induced product representation $\lambda_{ij} = \alpha_i \beta_j$ of z over the atoms $v_i - w_j$ has support $\text{supp } \alpha \times \text{supp } \beta$. With $g = -\nabla F(z) = q - p$, the *side scores* are $s_i^V = \langle g, v_i \rangle$ and $s_j^W = \langle p - q, w_j \rangle$, maintained incrementally from cached arrays.

Certificate. For any iterate, $\text{UB} = \|z\|$ and $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\| = (\min_i \langle z, v_i \rangle - \max_j \langle z, w_j \rangle) / \text{UB}$ satisfy $\text{LB} \leq \delta^* \leq \text{UB}$; the TA stops when the certified relative gap $(\text{UB} - \text{LB}) / \text{UB}$ is at most ε . The L_2 soft margin with regularisation parameter C , the ν -SVM and the kernel variants reduce to (1) exactly (augmented coordinates that turn the Gram matrix K into $K + I/C$; reduced convex hulls; feature-space images), see Appendix B.

Gaps. $g^{\text{FW}} = \max_{u \in Z} \langle g, u \rangle - \langle g, z \rangle$, $g^{\text{A}} = \langle g, z \rangle - \min_{u \in \text{supp } \lambda} \langle g, u \rangle$, $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}}$. Because $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, each gap splits into a V -side and a W -side term computable from the score arrays in $O(n + m)$ (Lemma 7), written $g_V^{\text{PW}}, g_W^{\text{PW}}$ (likewise for FW and A), with g_{side} the term of the side being stepped; a transfer $v_r - v_u$ on the V side is a pairwise direction of Z (Lemma 8).

3. Guarded block transfers

Algorithm 1 amortises one $O(n)$ scan over up to k transfers. Only the scan is amortised: a new receiver still costs one Gram or kernel column, $O(nd)$. Every candidate gain is available from cached quantities at a cost independent of n : with $S = \langle g, d \rangle = \sum_j \gamma_j \text{gap}_j$ and $D = \|d\|^2 = \gamma^\top \Gamma \gamma$,

Algorithm 1: One iteration of the guarded block-transfer step (side V ; W symmetric)

top- k receivers by s^V ; worst- k donors in $\text{supp } \alpha$; pair best-with-worst: (r_j, u_j) , $j \leq k'$ ($k' \leq k$ pairs, limited by the support size); pair 1 is the MDM pair (r^*, u^*)
 $d_j \leftarrow v_{r_j} - v_{u_j}$; $\text{gap}_j \leftarrow s_{r_j}^V - s_{u_j}^V$; discard pairs with $\text{gap}_j \leq 0$ and receivers that are donors;
 $\gamma_j \leftarrow \min(\text{gap}_j / \|d_j\|^2, c_j)$ // $c_j = \alpha_{u_j}$: donor capacity

- 1 **if** pair 1 uncapped ($\gamma_1 = \text{gap}_1 / \|d_1\|^2$) [case (a)] **then**
- 2 $\Delta_B \leftarrow$ gain of exact line search $\theta \in [0, 1]$ along $d = \sum_j \gamma_j d_j$; $\Delta_1 \leftarrow$ gain of the MDM step
- 3 take the step with the larger gain // guard, $O(k'^2)$
- 4 **else**
- 5 **if** the away step on u^* is boundary-clipped [case (b)] **then**
- 6 take it // drop step: $\alpha_{u^*} \rightarrow 0$
- 7 **else**
- 8 take the best of {block, MDM, away(u^*), toward(r^*)}
- 9 **end**
- 10 **end**

Γ the $k' \times k'$ Gram matrix of the pair directions ($O(k'^2)$ cached dot products), the block gain is $\theta S - \theta^2 D / 2$ at $\theta = \min(S/D, 1)$; the MDM, away and toward gains are the usual one-dimensional expressions. The *gain* of a step is the decrease $h(z_t) - h(z_{t+1})$. The cache update for a block is one matrix–vector product with the net per-index coefficients; columns are computed on first use and cached, so support columns cost nothing after their first appearance. Feasibility is immediate, since donors lose at most $\theta \gamma_j \leq c_j$. We call an iteration *case (a)* if pair 1 is uncapped and *case (b)* otherwise; the away step of line 6, which drives α_{u^*} to zero, is a *drop* step (a capped block pair may also zero a donor, which only helps the counts below).

Safe screening. With $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$ and $v_{\min} = \arg \min_i \langle z, v_i \rangle$, a zero-weight point with $\langle z, v_i \rangle - \langle z, v_{\min} \rangle > r \|v_i - v_{\min}\|$ is outside the optimal support and is removed (symmetrically on W with the max). Because the optimum’s support survives, the reduced problem has the same optimum and all subsequent lower bounds remain valid.

4. Analysis

Full proofs are in Appendix A.

Lemma 1 (Block gain) *Let $b_j = \gamma_j \|d_j\|$ over the k' retained pairs, all with $\text{gap}_j > 0$. Then $S \geq \sum_j b_j^2$, $D \leq k' S$, and the realised block gain satisfies $\Delta_B \geq S / (2k') \geq \Delta_1 / (2k')$, Δ_1 the MDM gain.*

With the guard, every case-(a) iteration gains $\max(\Delta_B, \Delta_1) \geq \Delta_1$: the guarded step never makes less progress than pairwise FW.

Lemma 2 (Good-step progress) *Work on the side with the larger pairwise gap. On any non-drop iteration, gain $\geq \min((g^{\text{PW}})^2 / (32M^2), g^{\text{PW}} / 8)$.*

Theorem 3 (Global linear convergence) *Suppose the guarded block-transfer algorithm with away fallback is run for one side per iteration (the side with the larger pairwise gap), using any $k \geq 1$, from any vertex pair (initial support $s_0 \leq 2$). Let $h_t = h(z_t)$. Then every non-drop iteration satisfies $h_{t+1} \leq (1 - \rho)h_t$ with*

$$\rho = \frac{1}{16} (\delta/M)^2.$$

For any horizon T , among the first T iterations the number of drop steps is at most $kT_{\text{good}} + s_0$, where T_{good} is the number of non-drop iterations; consequently, $h_T \leq h_0 (1 - \rho)^{(T-s_0)/(k+1)}$. Furthermore, the two-sided schedule of the implementation (larger-gap side first, second side skipped after a drop) satisfies the same statement with the same ρ and exponent $(T - s_0)/(2k + 1)$ (Lemma 10).

The rate combines Lemma 2 with $g^{\text{PW}} \geq \delta\sqrt{2h}$ (Lemma 9, from the pyramidal width and 1-strong convexity) and $g^{\text{PW}} \geq g^{\text{FW}} \geq h$. Drop counting is the away-step argument: each drop removes one support index and non-drop steps add at most k . There is no swap-step bound because case (b) never takes a capacity-clipped pairwise step. The implementation uses the two-sided schedule of Lemma 10; the batteries of §5 used the earlier unconditional V -then- W order, which differs only on drop iterations and agrees with it on the instances of Appendix C.

Proposition 4 (Near-orthogonal speed-up) *If all k' pairs are uncapped, the block line search is interior, and $|\langle d_i, d_j \rangle| \leq \eta \|d_i\| \|d_j\|$ for $i \neq j$, then the block gain is at least $(\sum_j \Delta_j)/(1 + \eta(k' - 1))$, where Δ_j is pair j 's individual gain.*

As $\eta \rightarrow 0$ one scan realises the sum of all k pairwise gains.

Lemma 5 (The certified gap is the FW gap) *At any full scan, $\text{UB} - \text{LB} = g^{\text{FW}}(z)/\text{UB}$, and $g^{\text{FW}}(z) \leq 2h + R\sqrt{2h}$ with $R = \max_{u \in Z} \|u\|$, the radius of Z about the origin.*

Theorem 6 (Screening activation) *Assume $\delta^* > 0$. Let F_V and F_W be the optimal faces (the vertices that can carry weight at the optimum). Let $\tau_i = \langle z^*, v_i \rangle - \min_j \langle z^*, v_j \rangle$ be the score margin of v_i (and symmetrically on W), with $\tau > 0$ the minimal nonzero margin. Finally, let L be the larger class diameter.*

*A zero-weight point with margin τ_i is screened as soon as $r < \tau_i/(3L)$. Consequently, once $h_t < \tau^4 \delta^{*2}/(2592 R^4 L^4)$, every zero-weight point outside $F_V \cup F_W$ is removed permanently. Under Theorem 3, this condition holds for every iteration $t > s_0 + \frac{k+1}{\rho} \log(2592 R^4 L^4 h_0/(\tau^4 \delta^{*2}))$. Following this iteration, each subsequent scan costs $O(|F_V| + |F_W| + |\text{supp } \alpha| + |\text{supp } \beta|)$ instead of $O(n + m)$.*

In words: since $r^2 \leq 2R(\text{UB} - \text{LB})$, a point with margin τ_i is screened once the certified gap falls below $\tau_i^2/(18RL^2)$. We expect the τ -dependence to be unavoidable, but do not prove it. Lemmas 1, 5 and 10, Proposition 4 and Theorem 6 are verified numerically on random and adversarial ensembles at machine precision (Appendix E).

5. Experiments

Setup. All solvers are validated against an exact quadratic program (QP): distances agree to 10^{-9} or better for every variant. Stopping rule $\varepsilon = 10^{-3}$ on the certified relative gap unless stated; baselines use matched tolerances on the same objective (Appendix C). Synthetic instances follow

Table 1: Regime battery on real data, five seeds, 600 s per solver, mean $\pm$ 95% CI. Kernel cells (RBF with width parameter $1/d$) use a class-balanced subsample of n_k points. Same objective per row: LIN, linear hard margin (ETA with screening vs LIBSVM at $C=10^6$; our SMO in brackets); KHM, feature-space hard margin (K-ETA vs LIBSVM at $C=10^6$); KL2, kernel L_2 margin, $C=1$ (K-ETA vs kernel SMO). conv = seeds in which ETA certified; oracle = $O(nd)$ column (ETA) or row (SMO) evaluations, thousands.

cell	dataset	n_k / d	ETA (s)	baseline (s)	conv	dist. ETA / base	acc. ETA / base	oracle
LIN	gisette	6000 / 5000	56.6 ± 5.9	51.4 ± 13.1 [405 $\pm$ 62]	5/5	3.6638 / 3.6640	.981 / .981	1.5 [11.1]
KHM	gisette	6000 / 5000	86.2 ± 14.8	104.6 ± 20.2	5/5	.05215 / .05215	.979 / .979	2.6
KHM	ijcnn1	9853 / 22	421 ± 149	1.3 ± 0.3	4/5	.00430 / .00430	.946 / .946	2.5
KHM	a9a	10000 / 123	> 600	26.4 ± 2.9	0/5	$\leq .00009$ / .0033	.690 / .752	5.9
KHM	covtype	10000 / 54	> 600	144 ± 7 (maxiter)	0/5	$\leq .00053$ / .00025	.704 / .712	7.2
KHM	w8a	6479 / 300	101 ± 159	—	5/5	hulls intersect	—	1.9
KL2	gisette	6000 / 5000	131 ± 34	420 ± 49	5/5	.06882 / .06882	.971 / .971	3.6 / 15.2
KL2	w8a	6479 / 300	11.5 ± 2.6	41.7 ± 10.0	5/5	.05682 / .05682	.975 / .975	3.6 / 20.2
KL2	a9a	10000 / 123	52.8 ± 71.0	57.7 ± 23.2	5/5	.03019 / .03019	.795 / .795	7.6 / 43.6
KL2	covtype	10000 / 54	26.0 ± 4.2	17.6 ± 0.5	5/5	.02609 / .02609	.792 / .792	9.1 / 56.2
KL2	ijcnn1	9853 / 22	9.2 ± 2.5	6.1 ± 0.7	5/5	.04790 / .04790	.959 / .959	4.6 / 26.1

[2] (two Gaussian clouds, $n = 5,000$ per class). Real data are the LIBSVM sets a9a, w8a, ijcn1, gisette and covtype (100k subsample), standardised, official test sets where they exist, five seeds, 600 s per solver and cell, all on one dedicated 32-vCPU Xeon 2.6 GHz machine (Appendix C). Every ETA-versus-baseline pair solves the *same* objective; “our SMO” is our own implementation and K-ETA the kernelised ETA.

Blocks and screening (Appendix C). Blocks cut iterations $4\text{--}30\times$ on synthetic and $6\text{--}9\times$ on real data relative to single MDM (plain pairwise FW), but wall-clock only where the $O(n)$ scan dominates rather than the columns (Tables 7, 6). The fixed $k=16$ is not tuned: ρ is k -independent, the block is self-limiting ($k' \leq |\text{supp } \alpha|$), non-positive gaps discarded, MDM fallback under the guard, so an oversized k costs $O(k^2)$ cached dot products and never a worse step), and $k=4$ and 16 agree within CI at $d=22$ and $d=5,000$. Screening gives $1.4\text{--}2.4\times$ at $\varepsilon = 10^{-5}$ and is neutral at 10^{-3} , as Theorem 6 predicts.

Real data, sparse-support regime (Table 1). TA I first certifies separability: gisette is separable, the other four are not (certified in 0.2–42 s), a diagnostic no baseline produces. On gisette ($d=5,000$; 1,233 of 6,000 points in the support) ETA ties LIBSVM on the linear hard margin, both $7\times$ ahead of our SMO; K-ETA is $1.2\times$ faster than LIBSVM on the RBF hard margin and $3.2\times$ faster than kernel SMO on the kernel L_2 margin, with identical solutions. Across all five KL2 cells K-ETA matches kernel SMO’s solution with $4\text{--}6\times$ fewer kernel-column evaluations; in wall-clock it is $3\text{--}4\times$ faster where columns are expensive (gisette, w8a) and $1.5\times$ slower where they are cheap (ijcnn1, covtype).

The KHM column shows where the geometry turns against the method. On ijcn1 both solvers return the identical solution (819 support vectors, accuracy 0.946), but K-ETA needs $1.5\text{--}2.5 \times 10^5$ iterations and 420 s against LIBSVM’s 1.3 s: the feature-space hulls are 0.004 apart in a space of diameter $\approx \sqrt{2}$, so δ/M in Theorem 3 is of order 10^{-3} , and LIBSVM’s second-order working-set selection suits that conditioning far better. On a9a and covtype the hulls are within 10^{-4} and

5×10^{-4} of touching: K-ETA times out at gap 3–10 while LIBSVM’s $C=10^6$ surrogate stops at its KKT tolerance 6–30 $\times$ farther out, so neither certifies anything; K-ETA’s closer iterate has the lower accuracy on a9a because the bisector direction is ill-determined as $\|p - q\| \rightarrow 0$. On w8a the feature-space hulls intersect (duplicate points with opposite labels); K-ETA certifies this in 101 s.

Real data, L_2 soft margins (Table 2, appendix). The same boundary appears within one problem class. Gisette is separable, so its L_2 margin has a sparse support and ETA converges to the certified gap in 30–45 s at every C , $7\times$ faster than our SMO. LIBLINEAR run to the same primal accuracy (tolerance 10^{-6}) takes 8 min to 1.5 h, but that is the tolerance needed to match ETA’s primal: at its default 10^{-4} it takes 5–66 s, comparable to ETA, and stops 5–24% above the certified optimum with lower test accuracy (Table 4). We claim no wall-clock advantage over LIBLINEAR on gisette; what ETA adds is the certificate, which reveals that the fast answer is not the optimum (Table 5). On the four overlapping datasets both geometric solvers hit the cap on every cell while LIBLINEAR finishes in 0.3–39 s. At $C=0.1$ ETA’s iterate already has LIBLINEAR’s test accuracy at the cap with the certificate still at 0.01–0.6; uncapped, a9a reaches 1% of the optimum after 811 s and certifies at 3,030 s, against LIBLINEAR’s 4 s (Table 5). At $C \geq 1$ the augmented hulls approach each other and the iterate at the cap is not usable; on covtype it is not usable at any C . The uncapped traces (Figure 1, appendix) show the linear-rate signature of Theorem 3 with 60–65% of the points in the support: the active atoms form a near-regular simplex on $n_s \approx 0.6n$ vertices, whose pyramidal width is $2/\sqrt{n_s}$ times its side [3], so $\rho \sim 1/n_s$, iterations grow linearly in n and, at $O(n)$ each, the cost is $O(n^2)$ -type.

6. Related work

Distance duality and the TA [1, 2]; MDM [4, 24], Gilbert [5] and Wolfe [25]; FW for polytope distance and SVMs [20–23]; the nearest-point and reduced-hull view [6, 8]; away steps [7], the pyramidal-width analysis [3] and the facial distance [26]. Swap-step-free pairwise variants [12, 13] and block-coordinate FW [9] are the closest algorithmic relatives; gap-safe screening [14, 16, 17] and active-set identification [18, 19] the closest to Theorem 6, with [10, 11, 15] as its SVM ancestors (Appendix D).

7. Conclusion

Guarded block transfers give the Triangle Algorithm a linear rate with explicit constants and no combinatorial swap-step bounds, together with a screening rule that is both safe and provably effective. The experiments locate the boundary the analysis predicts: sparse support and well-separated hulls, where the solver matches or beats LIBSVM and kernel SMO and supplies a certificate, against dense support or nearly touching hulls, where δ/M is small and primal coordinate descent or second-order SMO is faster by one to two orders of magnitude. Limitations: screening pays only when the certificate closes; the implementation is NumPy-level without sparse support; kernel experiments use 10k-point subsamples; the ν -SVM variant is not analysed; the constants are loose; and k is fixed (returns diminish beyond $k=4$, Table 6; an adaptive k , e.g. tied to the support size, is future work).

References

- [1] B. Kalantari. A characterization theorem and an algorithm for a convex hull problem. *Annals of Operations Research*, 226(1):301–349, 2015.
- [2] M. Gupta and B. Kalantari. An enhanced triangle algorithm for large-scale support vector machine optimization. *Journal of Intelligent Decision Making and Information Science*, 3(9s), 2026.
- [3] S. Lacoste-Julien and M. Jaggi. On the global linear convergence of Frank–Wolfe optimization variants. *NeurIPS*, 2015.
- [4] B. F. Mitchell, V. F. Demyanov, and V. N. Malozemov. Finding the point of a polyhedron closest to the origin. *SIAM J. Control*, 12(1):19–26, 1974.
- [5] E. G. Gilbert. An iterative procedure for computing the minimum of a quadratic form on a convex set. *SIAM J. Control*, 4(1):61–80, 1966.
- [6] S. S. Keerthi, S. K. Shevade, C. Bhattacharyya, and K. R. K. Murthy. A fast iterative nearest point algorithm for support vector machine classifier design. *IEEE Trans. Neural Networks*, 11(1):124–136, 2000.
- [7] J. Guélat and P. Marcotte. Some comments on Wolfe’s ‘away step’. *Mathematical Programming*, 35(1):110–119, 1986.
- [8] K. P. Bennett and E. J. Brendensteiner. Duality and geometry in SVM classifiers. *ICML*, 2000.
- [9] S. Lacoste-Julien, M. Jaggi, M. Schmidt, and P. Pletscher. Block-coordinate Frank–Wolfe optimization for structural SVMs. *ICML*, 2013.
- [10] L. El Ghaoui, V. Viallon, and T. Rabbani. Safe feature elimination in sparse supervised learning. *Pacific J. Optimization*, 8(4):667–698, 2012.
- [11] K. Ogawa, Y. Suzuki, and I. Takeuchi. Safe screening of non-support vectors in pathwise SVM computation. *ICML*, 2013.
- [12] K. Tsuji, K. Tanaka, and S. Pokutta. Pairwise conditional gradients without swap steps and sparser kernel herding. *ICML*, 2022.
- [13] F. Rinaldi and D. Zeffiro. Avoiding bad steps in Frank–Wolfe variants. *Computational Optimization and Applications*, 84(1):225–264, 2023.
- [14] O. Fercoq, A. Gramfort, and J. Salmon. Mind the duality gap: safer rules for the Lasso. *ICML*, 2015.
- [15] J. Wang, P. Wonka, and J. Ye. Scaling SVM and least absolute deviations via exact data reduction. *ICML*, 2014.
- [16] E. Ndiaye, O. Fercoq, A. Gramfort, and J. Salmon. Gap safe screening rules for sparsity enforcing penalties. *JMLR*, 18(128):1–33, 2017.

- [17] A. Shibagaki, M. Karasuyama, K. Hatano, and I. Takeuchi. Simultaneous safe screening of features and samples in doubly sparse modeling. *ICML*, 2016.
- [18] I. M. Bomze, F. Rinaldi, and D. Zeffiro. Active set complexity of the away-step Frank–Wolfe algorithm. *SIAM J. Optim.*, 30(3):2470–2500, 2020.
- [19] D. Garber. Revisiting Frank–Wolfe for polytopes: strict complementarity and sparsity. *NeurIPS*, 2020.
- [20] B. Gärtner and M. Jaggi. Coresets for polytope distance. *SoCG*, 2009.
- [21] K. L. Clarkson. Coresets, sparse greedy approximation, and the Frank–Wolfe algorithm. *ACM Trans. Algorithms*, 6(4):63, 2010.
- [22] I. W. Tsang, J. T. Kwok, and P.-M. Cheung. Core vector machines: fast SVM training on very large data sets. *JMLR*, 6:363–392, 2005.
- [23] R. Nanculef, E. Frandi, C. Sartori, and H. Allende. A novel Frank–Wolfe algorithm. Analysis and applications to large-scale SVM training. *Information Sciences*, 285:66–99, 2014.
- [24] J. López and J. R. Dorronsoro. Linear convergence rate for the MDM algorithm for the nearest point problem. *Pattern Recognition*, 48(4):1510–1522, 2015.
- [25] P. Wolfe. Finding the nearest point in a polytope. *Mathematical Programming*, 11(1):128–149, 1976.
- [26] J. Peña and D. Rodríguez. Polytope conditioning and linear convergence of the Frank–Wolfe algorithm. *Mathematics of Operations Research*, 44(1):1–18, 2019.

AI assistance. AI tools were used in the preparation of the code, experiments and manuscript; the author verified all results and is responsible for the content.

Appendix A. Proofs

Lemma 7 (Separability) *With the product representation, $g^{\text{PW}} = g_V^{\text{PW}} + g_W^{\text{PW}}$ where $g_V^{\text{PW}} = \max_i s_i^V - \min_{i \in \text{supp } \alpha} s_i^V \geq 0$ and g_W^{PW} likewise; the same holds for g^{FW} and g^{A} .*

Proof. $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, so the max over atoms and the min over the support separate; each side term is nonnegative since the support max dominates the support min and the global max dominates the support max. ■

Lemma 8 (Side transfers are Z -pairwise directions) *For $\alpha_u > 0$ and any $b \in \text{supp } \beta$, $v_r - v_u = (v_r - w_b) - (v_u - w_b)$ is a difference of an atom of Z and an atom in the support, and the transfer $\gamma \leq \alpha_u$ realises $z \leftarrow z + \gamma(v_r - v_u)$ inside Z .*

Proof of Lemma 1. Since $\gamma_j \leq \text{gap}_j / \|d_j\|^2$, we have $\gamma_j \text{gap}_j \geq \gamma_j^2 \|d_j\|^2 = b_j^2$. Consequently, $S = \sum_j \gamma_j \text{gap}_j \geq \sum_j b_j^2$. By the triangle inequality, $\|d\| \leq \sum_j \gamma_j \|d_j\| = \sum_j b_j$, and by Cauchy–Schwarz, $(\sum_j b_j)^2 \leq k' \sum_j b_j^2 \leq k' S$; hence $D \leq k' S$. Exact line search on the 1-smooth F gives $\Delta_B = S^2/(2D) \geq S/(2k')$ in the interior case and $\Delta_B = S - D/2 \geq S/2$ at the boundary $\theta = 1$, where $D \leq S$. Finally, $S \geq \gamma_1 \text{gap}_1 \geq \Delta_1$, since all terms are nonnegative and $\Delta_1 = \gamma_1 \text{gap}_1 - \gamma_1^2 \|d_1\|^2/2$. ■

Proof of Lemma 2. Let $g_S = \max(g_V^{\text{PW}}, g_W^{\text{PW}}) \geq g^{\text{PW}}/2$. Every direction below is a difference of two points of one class hull, so its norm is at most $\text{diam}(\text{conv } V) \leq M$. For a feasible direction with directional derivative $\sigma > 0$, squared norm D and maximal step $\bar{\theta}$, exact line search on a 1-smooth function gives gain $\geq \min(\sigma^2/(2D), \bar{\theta}\sigma/2)$ (interior optimum, or the clipped step evaluated with $\bar{\theta}D \leq \sigma$). In case (a), the guard’s gain is at least the MDM gain, whose derivative is g_S and whose interior optimum is attained, yielding gain $\geq g_S^2/(2M^2) \geq (g^{\text{PW}})^2/(8M^2)$. In case (b), non-drop, $g_S = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$, so one term is at least $g_S/2 \geq g^{\text{PW}}/4$. If it is the FW term, the toward step (maximal step 1) gives gain $\geq \min((g_{\text{side}}^{\text{FW}})^2/(2M^2), g_{\text{side}}^{\text{FW}}/2) \geq \min((g^{\text{PW}})^2/(32M^2), g^{\text{PW}}/8)$; if it is the away term, the away line search is interior (non-drop) and gain $\geq (g_{\text{side}}^{\text{A}})^2/(2M^2) \geq (g^{\text{PW}})^2/(32M^2)$. The algorithm takes the best available candidate. ■

Lemma 9 (Geometric strong convexity; [3, Thm. 6] with $\mu = 1$) *For every $z \in Z$ with a representation whose support contains an away atom, $g^{\text{PW}}(z) \geq \delta\sqrt{2h(z)}$ and $g^{\text{PW}}(z) \geq g^{\text{FW}}(z) \geq h(z)$.*

Proof. This is Theorem 6 of [3] for a 1-strongly convex objective; we include the short argument. By the definition of the pyramidal width [3, Def. 3], for $e = (z^* - z)/\|z^* - z\|$ the FW atom s and the away atom v satisfy $\langle g, s - v \rangle \geq \delta \langle g, e \rangle$. By 1-strong convexity, $\langle g, z^* - z \rangle \geq h + \frac{1}{2}\|z^* - z\|^2$, so with $\varrho = \|z^* - z\|$, $g^{\text{PW}} \geq \delta(h/\varrho + \varrho/2) \geq \delta\sqrt{2h}$ by AM–GM. Finally $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}} \geq g^{\text{FW}} = \max_u \langle g, u - z \rangle \geq \langle g, z^* - z \rangle \geq h$ by convexity. ■

Proof of Theorem 3. By Lemmas 2 and 9 a non-drop iteration gains at least

$$\min(2\delta^2 h/(32M^2), h/8) = (\delta/M)^2 h/16$$

using $\delta \leq M$; hence $h_{t+1} \leq (1 - \rho)h_t$. Drop counting: non-drop iterations add at most k support indices, drop iterations add none and remove at least one, and the support size stays ≥ 2 from s_0 ; so the number of removals, hence of drops, is at most $kT_{\text{good}} + s_0$, so that

$$T_{\text{good}} \geq (T - s_0)/(k + 1).$$

All steps, drops included, are descent steps, so h is monotone. ■

Lemma 10 (Two-sided schedule) *Let every iteration compute both side gaps, take the guarded step of Algorithm 1 on the side with the larger pairwise gap first, and take the other side’s guarded step only if the first step was not a drop; an iteration counts as a drop iff its first step is a drop. Then every non-drop iteration gains at least $\min((g^{\text{PW}})^2/(32M^2), g^{\text{PW}}/8)$ and the number of drops is at most $2kT_{\text{good}} + s_0$; hence Theorem 3 holds for this schedule with the same ρ and exponent $(T - s_0)/(2k + 1)$.*

Proof. On a non-drop iteration the first step is exactly the step analysed in Lemma 2, so it gains at least the stated amount; the second step is an exact line search along a feasible descent direction (or is skipped), so it cannot increase h . Drop counting: a drop iteration takes a single away step that removes one support index and adds none, because the second side is skipped; a non-drop iteration takes two guarded steps and adds at most k indices each. Removals, hence drops, therefore number at most $2kT_{\text{good}} + s_0$. ■

Remark 11 *The skip is what makes the count work. If the second side always steps, an iteration whose analysed side drops can add up to k indices on the other side, and the count becomes $T_{\text{drop}} \leq 2kT_{\text{good}} + kT_{\text{drop}} + s_0$, which is vacuous for $k \geq 1$.*

Proof of Proposition 4. $D = \sum_{i,j} \gamma_i \gamma_j \langle d_i, d_j \rangle \leq \sum_j b_j^2 + \eta \sum_{i \neq j} b_i b_j \leq (1 + \eta(k' - 1)) \sum_j b_j^2 \leq (1 + \eta(k' - 1))S$ by Cauchy–Schwarz on the cross terms; then $\Delta_B = S^2/(2D) \geq S/(2(1 + \eta(k' - 1)))$, and for uncapped pairs $S = \sum_j 2\Delta_j$. ■

Proof of Lemma 5. $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\|$ by definition of the supporting hyperplane bound and $\text{UB} = \|z\|$; subtracting, $\text{UB} - \text{LB} = (\langle z, z \rangle - \min_u \langle z, u \rangle) / \|z\| = g^{\text{FW}} / \text{UB}$. At the optimum $0 = \langle z^*, z^* \rangle - \min_u \langle z^*, u \rangle$; subtracting from $g^{\text{FW}}(z)$, $g^{\text{FW}}(z) = (\|z\|^2 - \|z^*\|^2) + (\min_u \langle z^*, u \rangle - \min_u \langle z, u \rangle) \leq 2h + \max_u \langle z^* - z, u \rangle \leq 2h + R\|z - z^*\| \leq 2h + R\sqrt{2h}$ by 1-strong convexity. ■

Proof of Theorem 6. Strong convexity gives $\|z - z^*\| \leq r$ for the screening radius $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$. Let $m^* = \min_j \langle z^*, v_j \rangle$ and pick $v' \in F_V$; since $v_{\min}$ minimises $\langle z, \cdot \rangle$, $\langle z^*, v_{\min} \rangle - m^* = \langle z, v_{\min} - v' \rangle + \langle z^* - z, v_{\min} - v' \rangle \leq rL$. Hence $\langle z, v_i - v_{\min} \rangle = \langle z^*, v_i - v_{\min} \rangle + \langle z - z^*, v_i - v_{\min} \rangle \geq \tau_i - rL - r\|v_i - v_{\min}\|$, and the rule fires when this exceeds $r\|v_i - v_{\min}\|$, for which $\tau_i > 3rL$ suffices since $\|v_i - v_{\min}\| \leq L$. For the activation time, $r^2 \leq 2R(\text{UB} - \text{LB})$ (for $\text{LB} \geq 0$, $r^2 = (\text{UB} - \text{LB})(\text{UB} + \text{LB})$ and $\text{UB} + \text{LB} \leq 2R$; for $\text{LB} < 0$, $r^2 = \text{UB}^2$ and $\text{UB} \leq 2(\text{UB} - \text{LB})$), and $2R(\text{UB} - \text{LB}) = 2Rg^{\text{FW}}/\text{UB} \leq (2R/\delta^*)(2h + R\sqrt{2h}) \leq (4R^2/\delta^*)\sqrt{2h}$ for $h \leq R^2/2$ (Lemma 5); requiring $r^2 \leq \tau^2/(9L^2)$ yields the stated condition on h_t , and Theorem 3 with $\log(1/(1 - \rho)) \geq \rho$ converts it into the iteration bound. Permanence follows from the safety of the rule. ■

Appendix B. Exact reductions

L_2 soft margin. The problem $\min \frac{1}{2}\|w\|^2 + \frac{C}{2} \sum_i \xi_i^2$ is the hard-margin problem on the augmented points $\tilde{x}_i = [x_i; e_i/\sqrt{C}]$, e_i the i -th standard basis vector, whose Gram matrix is $K + I/C$ (K the Gram matrix of the original points) and whose augmented hulls never intersect; let δ_C denote their distance. The augmentation is never materialised: the extra coordinates of an iterate are exactly its convex weights, so every augmented inner product is a cached original-space product plus a sparse diagonal correction, $\langle \tilde{p}, \tilde{v}_i \rangle = \langle p, v_i \rangle + \alpha_i/C$ and $\|\tilde{p}\|^2 = \|p\|^2 + \|\alpha\|^2/C$. The separator is recovered as $w = 2(p - q)/\delta_C^2$, $b = (\|\tilde{q}\|^2 - \|\tilde{p}\|^2)/\delta_C^2$, and the primal value is $P = 2/\delta_C^2$. Validation: the solver agrees with an exact QP on the explicitly augmented problem to 10^{-10} across random instances and values of C ; the regularisation path δ_C is monotone in C ; on a9a ($C=0.1$) its certified primal 687.07 equals LIBLINEAR’s to five digits.

ν -SVM. The L_1 (hinge) soft margin is the nearest-point problem between *reduced* convex hulls $R(V, \mu) = \{\sum a_i v_i : \sum a_i = 1, 0 \leq a_i \leq \mu\}$, $\mu = 2/(\nu\ell)$ with $\ell = n + m$ the number of training points [8]. Two changes: the extreme-point oracle becomes a capped top- k blend (weight μ on the $\lfloor 1/\mu \rfloor$ best points in the search direction, via a partial sort, still $O(n)$), and the primary step is the capped MDM transfer with blended toward steps as fallback; caching and the reduced support-function lower bound carry over unchanged. Validation: agreement with the exact box-constrained QP to 10^{-9} – 10^{-14} and with LIBSVM’s NuSVC to $\sim 10^{-10}$ through the mapping $\delta^* = \|w_{\text{svc}}\|/\varsigma$, w_{svc} LIBSVM’s weight vector and ς the per-class sum of the rescaled dual coefficients. The reduced hull is a polytope, so (1) still applies, but its vertices are blends and the capped transfer is not the pairwise step analysed in §4; the analysis of this variant is left open.

Kernels. The algorithm touches data only through inner products, so kernelisation changes plumbing, not algorithm: Gram columns become kernel columns $\kappa(V, x_i)$, squared norms become kernel diagonals, iterates exist purely as convex weights (the exact cache refresh is reconstructed from the support’s cached kernel columns), and the classifier is the kernel expansion of the witness-pair bisector. A ridge $K + I/C$ gives the kernel L_2 margin by the same diagonal trick. Validation: the linear kernel reproduces the Euclidean solver to 10^{-12} ; RBF distances match the exact kernel QP to 10^{-9} and LIBSVM’s hard-margin solution, via $\|w_{\mathcal{H}}\|^2 = (\alpha \circ y)^\top K(\alpha \circ y)$, $w_{\mathcal{H}}$ the feature-space normal, for LIBSVM’s dual coefficients α and labels y , to 10^{-5} ; kernel SMO matches the QP to 10^{-8} .

Appendix C. Experimental details

Solver settings and tolerance equivalence. ETA: block transfers with $k = 16$, guarded, away fallback, pairwise anti-zig-zag, stop at certified relative gap 10^{-3} , iteration cap 2×10^6 , 600 s cap. The batteries were run with the unconditional V -then- W order; the schedule of Lemma 10 (larger gap first, skip after a drop) differs from it only on drop iterations, and Table 3 shows the two agree on the synthetic instances of this appendix; screening (every 25 iterations) only in the LIN cell, where it never activated at 10^{-3} , as Theorem 6 predicts. Our SMO: maximal-violating-pair selection, KKT gap 10^{-3} , same caps; hard margin as $C = 10^{12}$; kernel SMO on $K + I/C$. LIBLINEAR (scikit-learn 1.8 `LinearSVC`): squared hinge with parameter $C/2$, so that its objective equals $\frac{1}{2}\|w\|^2 + \frac{C}{2} \sum \xi_i^2$; tolerance 10^{-6} , chosen so that its final primal matches ETA’s certified value on every cell where ETA converges (a9a 687.07 vs 687.07; gisette 0.14900 vs 0.14893, inside ETA’s certified 8×10^{-4} ; ijcn1 550.6 vs 550.5 at ETA gap 0.01); this tolerance is also why LIBLINEAR needs 18,000 iterations and over an hour on gisette at $C=1$, and 35,000 iterations and 1.5 h at $C=10$, where it nevertheless stops 5% above the certified optimum (0.1566 vs 0.1490). The gisette LIBLINEAR times in Table 2 are therefore the cost of matching ETA’s primal; Table 4 reports the same configuration at scikit-learn’s default tolerance 10^{-4} (five seeds, run separately without concurrent load): 5–66 s, but primals 5–24% above the certified optimum and lower test accuracy. Intercept scaling 100; iteration cap 2×10^5 . LIBSVM (scikit-learn `SVC`): tolerance 10^{-3} , 1000 MB cache, iteration cap 2×10^6 (reported as *maxiter* when hit); hard margins as $C = 10^6$; RBF kernel $\exp(-\|x - x'\|^2/d)$. Oracle counts: ETA counts Gram or kernel columns evaluated against a full class ($O(nd)$ or $O(n_k d)$ each); SMO counts kernel rows, the same unit. Columns are cached without eviction and the block Gram matrix Γ is assembled from four cached entries per pair, so peak cache memory is the column count times n_k doubles: 0.61 GB for the largest cell (a9a KL2, 7,600 columns) and 0.36 GB for ijcn1 KL2.

Data and seeds. a9a, w8a, ijcn1 and gisette use the LIBSVM training and test files; covtype (binary, scaled) is subsampled to 100,000 points and split 75/25 by seed. Features are standardised on the training set. Kernel cells use a class-balanced subsample of at most 5,000 points per class (w8a’s minority class has 1,479 points, hence $n_k = 6,479$; gisette uses all 6,000). Seeds vary the covtype split, the kernel subsamples and the timing environment; on datasets with fixed splits the deterministic solvers return identical solutions across seeds, which is why several accuracy CIs are zero. Confidence intervals are t -based 95% intervals over seeds.

ETA on real data (L_2 soft margin, $C = 0.1$): certified gap to the 10^{-3} stopping rule

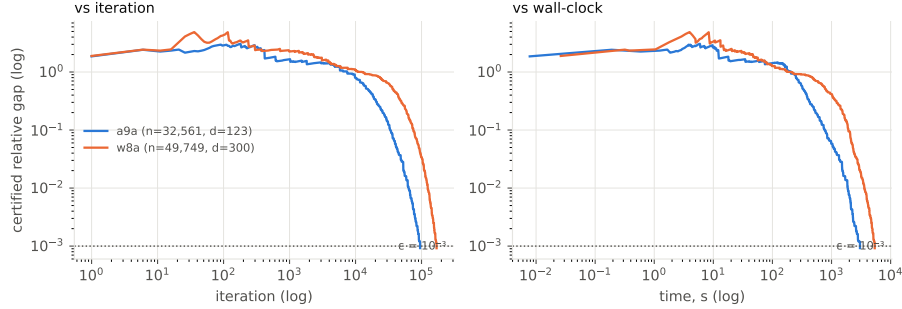


Figure 1: Certified relative gap of ETA on a9a and w8a (L_2 , $C = 0.1$, uncapped), versus iteration and wall-clock: linear-rate convergence in the dense-support regime, at $O(n^2)$ -type cost.

Hardware and concurrency. One dedicated 32-vCPU Intel Xeon 2.6 GHz Lightning Studio, 122 GB RAM, Linux, Python 3.12, NumPy 2.4 with OpenBLAS 0.3.31, `OMP_NUM_THREADS=1`; the L_2 battery ran 14 cells and the regime battery 12 cells concurrently (the gisette, covtype and w8a $C=10$ cells of Table 2 were completed in later passes with 11–14 workers and no other load; the runs of Tables 4 and 6 were made separately with no concurrent load, which is why their gisette $k=16$ time is below Table 2’s; the synthetic runs of Tables 3 and 7 likewise, and the five trials of Figure 3 ran as seven concurrent single-threaded jobs). The uncapped a9a/w8a runs in Figure 1 were made earlier on the same machine under a different load (LIBLINEAR took 12 and 48 s there against 4 and 23 s in the capped battery), so ratios across the two runs carry a factor of roughly two of uncertainty; ratios within a battery do not.

Reference solver in the KL2 cells. The hinge-loss SVC (`rbf`, $C=1$), a different objective, took 8.6 ± 2.9 (a9a), 3.4 ± 0.3 (covtype), 83.5 ± 13.1 (gisette), 1.1 ± 0.0 (ijcn1) and 4.9 ± 1.2 s (w8a), with test accuracies 0.789, 0.783, 0.977, 0.950 and 0.975, within one point of the L_2 solutions and on either side of them.

Step ablation, blocks and shrinking (synthetic). Iteration counts at $\varepsilon = 10^{-6}$ (single instances): $d=30$, $n=200$: toward 158,941, midpoint 148,996, away 106, pairwise 181; $d=100$, $n=1,000$: toward and midpoint fail within 300,000, away 15,886, pairwise 3,921. At the working tolerance 10^{-3} the pairwise/away gain over the midpoint heuristic is only $\sim 11\%$, which is why the heuristic sufficed in [2]. Block transfers cut iterations 4–30 $\times$ relative to single MDM (e.g. 43,486 $\rightarrow$ 1,551 on a dense-support L_2 instance at $d=1,000$) and are the only configuration that reaches the certified toler-

Table 2: L_2 soft-margin battery: five seeds, 600 s per solver for ETA and our SMO (our own implementation; LIBSVM does not solve this objective), LIBLINEAR at tolerance 10^{-6} (Appendix C) and unbounded; mean $\pm$ 95% CI, which on the fixed-split datasets (all but covtype) measures timing noise only. “cap” = timed out in every seed; ETA’s certified relative gap is reported at its stopping point (mean [min, max]); accuracies are those of the returned iterate.

dataset (n / d)	C	ETA (s)	SMO (s)	LIBLINEAR (s)	ETA gap at stop	acc. ETA / SMO / LIBLINEAR
gisette (6,000 / 5,000)	0.1	35.3 ± 6.8	243 ± 53	500 ± 41	8×10^{-4}	.981 / .981 / .981
	1	43.3 ± 5.3	295 ± 6	3916 ± 115	8×10^{-4}	.981 / .981 / .981
	10	33.2 ± 4.6	236 ± 36	5396 ± 197	10×10^{-4}	.981 / .981 / .977
a9a (32,561 / 123)	0.1	cap	cap	3.7 ± 0.8	0.15 [0.10, 0.23]	.849 / .845 / .849
	1	cap	cap	6.8 ± 1.7	1.26 [1.25, 1.28]	.810 / .780 / .849
	10	cap	cap	2.3 ± 0.5	1.83 [1.80, 1.86]	.635 / .763 / .849
ijcnn1 (49,990 / 22)	0.1	cap	478 ± 181	0.3 ± 0.1	0.010 [0.005, 0.016]	.918 / .918 / .918
	1	cap	cap	0.6 ± 0.2	0.68 [0.61, 0.73]	.901 / .917 / .918
	10	cap	cap	0.4 ± 0.1	1.34 [1.27, 1.40]	.656 / .852 / .918
w8a (49,749 / 300)	0.1	cap	cap	22.7 ± 4.3	0.57 [0.50, 0.66]	.987 / .985 / .987
	1	cap	cap	20.6 ± 6.6	1.17 [1.14, 1.21]	.923 / .979 / .987
	10	cap	cap	13.2 ± 1.1	2.02 [1.94, 2.14]	.720 / .971 / .987
covtype (100,000 / 54)	0.1	cap	cap	19.4 ± 15.7	1.04 [1.02, 1.05]	.568 / .660 / .756
	1	cap	cap	6.1 ± 3.6	1.42 [1.23, 1.73]	.508 / .541 / .756
	10	cap	cap	5.0 ± 1.8	2.01 [1.20, 2.37]	.517 / .514 / .756

ance on MNIST odd-vs-even. Figure 2 shows the screening speed-ups and Figure 3 the consolidated comparison, five trials per class (mean $\pm$ 95% CI): on the four hard-margin classes the optimised solver is $1.2\text{--}1.3\times$ faster than LIBSVM on average, with overlapping intervals (0.73 ± 0.11 vs 0.88 ± 0.21 s at $d=1,000$; 18.6 ± 1.9 vs 22.5 ± 4.2 s at $d=10,000$; 0.74 ± 0.17 vs 0.88 ± 0.30 s at $\varepsilon=10^{-5}$, where the paper-style configuration fails to converge in every trial; RBF 2.44 ± 0.34 vs 3.14 ± 0.71 s); on the synthetic L_2 margin ($d=1,000$, $C=1$) it ties LIBLINEAR (19.6 ± 7.4 vs 23.5 ± 11.2 s); on MNIST odd-vs-even it is $2.7\times$ slower (176 ± 27 vs 66 ± 26 s; single MDM never converges) and on the ν -SVM $1.7\times$ slower (65.3 ± 3.7 vs NuSVC’s 37.4 ± 1.4 s). Distances agree within tolerance on every row. An earlier run of the same script on a shared development container had shown the optimised solver $1.6\text{--}2.5\times$ ahead on six of the seven classes; that machine’s Python-to-C timing ratio did not reproduce on dedicated hardware, and only the dedicated numbers are reported here.

Table 6 was produced by the current code, whose default is the schedule of Lemma 10; its distances and accuracies coincide with those of Tables 1 and 2, which used the unconditional order, so the two schedules agree on real data as on the synthetic instances of Table 3. Away fallbacks that drop an index occur on 1–36% of block iterations and never with single transfers, and their share grows with k : larger blocks add more indices per iteration, and the drops remove them.

Appendix D. Extended related work

Swap steps. Blended pairwise conditional gradients [12] also remove pairwise FW’s swap-step bound with a guard, there between the pairwise step and a FW step, and [13] avoid bad steps with a short-step chain. Our guard compares an aggregated block of k transfers with the single MDM step and falls back to an away step, so the drop-step count of away-step FW applies unchanged.

Table 3: Two-sided schedules on the synthetic instances of this appendix ($n=5,000$ per class, $k=16$): the unconditional V -then- W order used for the batteries versus the schedule of Lemma 10 (larger-gap side first, second side skipped after a drop). Drop steps and skipped second-side steps are counted over the run.

instance	schedule	time (s)	iterations	drops	skips	distance
hard margin, $d=1000$, $\varepsilon=10^{-5}$	unconditional	0.1	16	0	0	55.339365
hard margin, $d=1000$, $\varepsilon=10^{-5}$	Lemma 10	0.2	16	0	0	55.339365
L_2 overlap, $d=100$, $C=1$	unconditional	7.9	11,216	638	0	0.078400
L_2 overlap, $d=100$, $C=1$	Lemma 10	8.1	10,701	599	316	0.078400
L_2 overlap, $d=1000$, $C=1$	unconditional	4.3	2,951	172	0	0.578143
L_2 overlap, $d=1000$, $C=1$	Lemma 10	4.3	3,001	180	121	0.578143

Table 4: LIBLINEAR on gisette at its default tolerance (10^{-4}) versus the matched tolerance (10^{-6}) of Table 2; five seeds. ETA’s certified primal is 0.14835 / 0.14893 / 0.14899 at $C = 0.1 / 1 / 10$, reached in 30–45 s with test accuracy 0.981.

C	tol	time (s)	iterations	primal (vs certified)	accuracy
0.1	10^{-4}	66 ± 10	363	0.15648 (+5.5%)	.978
0.1	10^{-6}	500 ± 41	2,434	0.14835 (+0.0%)	.981
1	10^{-4}	5.5 ± 0.8	10	0.18427 (+24%)	.974
1	10^{-6}	3916 ± 115	18,063	0.14900 (+0.05%)	.981
10	10^{-4}	4.8 ± 0.1	8	0.18487 (+24%)	.973
10	10^{-6}	5396 ± 197	35,411	0.15663 (+5.1%)	.977

Block-coordinate FW [9] blocks over *coordinates* of a product domain rather than aggregating pairs on one polytope.

Screening and identification. Gap-safe rules [14, 16, 17] screen with a radius from the duality gap and identify the support in finite time; the active-set complexity of away-step FW [18] and the strict-complementarity bounds of [19] give the same identification from the algorithm’s side. Theorem 6 is the polytope-distance instance, with constants tied to the TA certificate; the SVM rules of [10, 11] and the exact data reduction of [15] are its ancestors.

Relation to other FW variants. The toward, away, pairwise and block rows of the step ablation in Appendix C are Frank–Wolfe, away-step FW, pairwise FW and our block step on the same polytope and objective; Wolfe’s minimum-norm-point method [25] and blended pairwise conditional gradients (BPCG) [12] were not run. BPCG’s blending rule compares the away and FW gaps and its linear rate has the same pyramidal-width form as Theorem 3; our case (b) instead compares the realised gains of the block, MDM, away and toward candidates, so each step is at least as good as the best single candidate, and it is the away fallback rather than the blending that yields the drop-step count. The facial distance of [26] is a computable alternative to the pyramidal width; a bound on δ for the difference atom set $A = \{v_i - w_j\}$ in terms of the two class hulls is open.

Synthetic results in brief. Away and pairwise steps converge at $\varepsilon = 10^{-6}$ where the toward and midpoint steps of [2] do not. Screening reduces 10,000 points to 36–1,054 survivors at $\varepsilon = 10^{-5}$.

Table 5: Time to 1% relative primal suboptimality, seed-0 traces. For ETA the certified dual value $2/\text{UB}_t^2$ increases to the optimum, so $s_t = 1 - (2/\text{UB}_t^2)/P^*$ with P^* the best known primal (the lower of LIBLINEAR’s 10^{-6} value and ETA’s certified value); “certified 1%” is the first time the certificate itself proves $1 - \text{LB}_t^2/\text{UB}_t^2 \leq 1\%$. ETA’s a9a and w8a times are from the uncapped runs of Figure 1. For LIBLINEAR the entry is the time of the coarsest tolerance whose primal is within 1% of P^* ; where only 10^{-6} has been run it is an upper bound. On the remaining cells ETA is not within 1% at the 600 s cap ($s = 30\text{--}99\%$).

dataset	C	P^*	ETA 1%	ETA 0.1%	cert. 1%	cert. 10^{-3}	LIBLINEAR 1%
gisette	0.1	0.14835	40 s	42 s	43 s	43 s	≤ 500 s (tol 10^{-6})
gisette	1	0.14893	44 s	46 s	47 s	48 s	≤ 3916 s (tol 10^{-6})
gisette	10	0.14899	34 s	36 s	37 s	37 s	none at 10^{-4} or 10^{-6}
a9a	0.1	687.07	811 s	1,413 s	2,121 s	3,030 s	4 s
w8a	0.1	108.80	2,084 s	2,960 s	4,162 s	5,386 s	23 s
ijcnn1	0.1	550.61	301 s	472 s	> 600 s	> 600 s ($s=0.05\%$)	0.3 s

Table 6: Block-size ablation on real data (ETA only, three seeds, run separately without concurrent load): the gisette L_2 margin ($C=1$, $d=5,000$) and the ijcnn1 kernel L_2 cell of Table 1 ($d=22$, $n_k=9,853$). $k=1$ is plain pairwise FW (single MDM transfers, no block, no guard); drops = away fallbacks that removed a support index. Every cell converged to the certified gap with identical distance and accuracy per dataset.

instance	k	time (s)	iterations	columns	drops
gisette L_2 , $C=1$	1	26.3 ± 5.4	4,946	1,300	0
gisette L_2 , $C=1$	4	25.3 ± 5.1	1,071	1,349	80
gisette L_2 , $C=1$	16	27.8 ± 4.4	831	1,480	301
ijcnn1 kernel L_2 , $C=1$	1	13.6 ± 0.4	12,796	4,520	0
ijcnn1 kernel L_2 , $C=1$	4	6.6 ± 0.5	2,358	4,522	30
ijcnn1 kernel L_2 , $C=1$	16	5.8 ± 0.3	1,396	4,590	98

On seven synthetic and small-real problem classes (Figure 3, five trials each) the optimised solver is $1.2\text{--}1.3\times$ faster than LIBSVM on the four hard-margin classes (intervals overlap), ties LIBLINEAR on the synthetic L_2 margin, and is $1.7\text{--}2.7\times$ slower than NuSVC and LIBLINEAR on the ν -SVM and MNIST classes. On the four classes where single MDM also converges, blocks and guard change wall-clock by $1.0\text{--}1.3\times$; their clear wins are the two classes (tight tolerance, MNIST) where single MDM does not converge at all.

Appendix E. Numerical verification of the analysis

Lemma 1 was checked on 4,000 random ensembles including heavily clipped and boundary cases (worst slack at machine precision); Proposition 4 on 500 near-orthogonal ensembles; Lemma 5 to 10^{-9} at arbitrary iterates; Lemma 10 on instrumented two-sided runs (no drop iteration added a support index, and every non-drop iteration met the gain bound with M replaced by the computable upper bound $\text{diam } V + \text{diam } W$); and Theorem 6 on instances whose optimal face was computed by an exact QP: at every activating round all zero-weight non-face points were screened, and the

Table 7: Block-size ablation on the synthetic instances (current code): single MDM ($k=1$) versus blocks of k transfers. Iterations fall with k ; wall-clock does not fall in proportion, because each block iteration touches more columns.

instance	k	time (s)	iterations	columns	drops
hard margin, $d=1000$, $\varepsilon=10^{-5}$	1	0.3	296	104	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	4	0.2	36	103	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	16	0.2	16	103	0
hard margin, $d=1000$, $\varepsilon=10^{-5}$	32	0.2	16	109	0
L_2 overlap, $d=100$, $C=1$	1	10.7	75,021	1,343	0
L_2 overlap, $d=100$, $C=1$	4	11.2	24,606	1,632	416
L_2 overlap, $d=100$, $C=1$	16	9.0	10,701	2,404	599
L_2 overlap, $d=100$, $C=1$	32	10.4	7,606	3,076	305
L_2 overlap, $d=1000$, $C=1$	1	12.2	43,521	1,251	0
L_2 overlap, $d=1000$, $C=1$	4	6.7	10,071	1,330	177
L_2 overlap, $d=1000$, $C=1$	16	4.3	3,001	1,348	180
L_2 overlap, $d=1000$, $C=1$	32	4.6	1,741	1,396	170

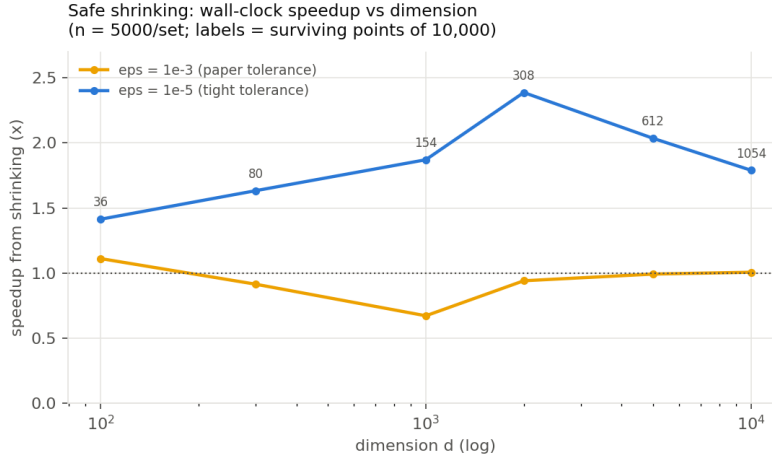


Figure 2: Safe screening: wall-clock speed-up vs dimension at two tolerances on the separated synthetic protocol; labels are surviving points of 10,000.

final working set equalled the optimal face exactly. Over 5,000 instrumented live block steps none increased the objective. All checks are test files in the released code.

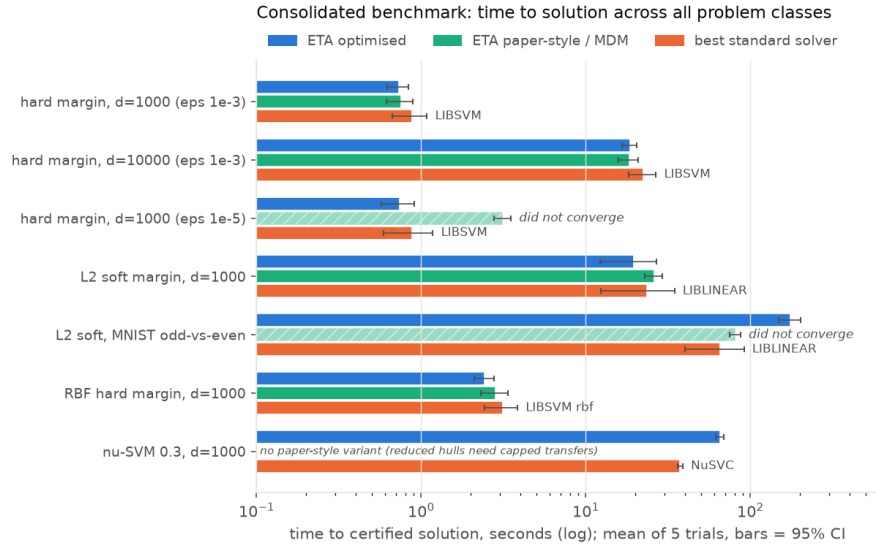


Figure 3: Time to certified solution on seven problem classes: optimised ETA vs the paper-style configuration vs the best standard solver per class; five trials per class, mean with 95% CI. The ν -SVM row has no paper-style bar because the toward-step algorithm of [2] has no reduced-hull variant (reduced hulls need capped transfers).